Winding Numbers, Pseudo-Anosov Structure, and the (ν_2, ν_3) Equilibrium in the Collatz Dynamical System

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1 Introduction and Motivation

The Collatz map $T : \mathbb{N} \to \mathbb{N}$ is defined by

$$T(n) = \begin{cases} n/2 & \text{if } n \equiv 0 \pmod 2, \\ 3n+1 & \text{if } n \equiv 1 \pmod 2. \end{cases} \tag{1}$$

The Collatz conjecture asserts that for every $n \geq 1$, the orbit $n, T(n), T^2(n), \ldots$ eventually reaches 1. Despite extensive numerical verification (up to $n \sim 10^{20}$, see [9]), the conjecture remains open.

These notes develop a topological perspective on Collatz dynamics by introducing *winding numbers* associated to trajectories. The central observation is that each trajectory from n to 1 accumulates a pair of integers $(\nu_2(n), \nu_3(n))$ counting even and odd steps respectively, and that this pair constitutes a *winding number* on a 2-torus $\mathbb{T}^2$ equipped with a distinguished irrational foliation of slope $\log_2 3$. The deviation of the empirical ratio $\rho(n) = \nu_2/\nu_3$ from the equilibrium value $\log_2 3$ encodes dynamically meaningful information that is invisible in the raw orbit data.

We establish an exact algebraic identity (Proposition 2.4) that decomposes ρ into a universal equilibrium term plus explicit correction terms, prove that $\rho > \log_2 3$ for all terminating trajectories (Corollary 2.5), and use eigenbasis coordinates adapted to the equilibrium foliation to reveal a pseudo-Anosov structure in the dynamics. Computational experiments on 10^6 trajectories confirm the theoretical predictions and expose additional arithmetic structure in the torus residues $(\nu_2 \bmod k, \nu_3 \bmod k)$.

2 The Winding Number Pair

2.1 Definitions

Definition 2.1 (Parity counts and winding pair)**.** For $n \geq 2$ with finite Collatz trajectory $n = a_0, a_1, \ldots, a_k = 1$, define:

$$\nu_2(n) := \#\{i : 0 \leq i < k,\ a_i \text{ even}\}, \tag{2}$$
$$\nu_3(n) := \#\{i : 0 \leq i < k,\ a_i \text{ odd}\}. \tag{3}$$

The *winding pair* of n is $w(n) := (\nu_2(n), \nu_3(n)) \in \mathbb{Z}^2_{\geq 0}$, and the *winding ratio* is

$$\rho(n) := \frac{\nu_2(n)}{\nu_3(n)}. \tag{4}$$

The total stopping time is $\sigma(n) := \nu_2(n) + \nu_3(n)$.

The terminology "winding number" is justified by the following construction.

Definition 2.2 (Torus winding)**.** Consider the 2-torus $\mathbb{T}^2 = (\mathbb{R}/\mathbb{Z})^2$. The Collatz trajectory of n determines a path $\gamma_n : \{0, 1, \ldots, \sigma(n)\} \to \mathbb{T}^2$ starting at the origin, where

- each even step increments the first coordinate (mod 1),
- each odd step increments the second coordinate (mod 1).

The winding pair $w(n)$ is the element of $H_1(\mathbb{T}^2; \mathbb{Z}) \cong \mathbb{Z}^2$ represented by γ_n.

Remark 2.3. The first homology group $H_1(\mathbb{T}^2; \mathbb{Z}) \cong \mathbb{Z}^2$ provides two independent integer-valued winding numbers. The ratio ρ is the slope of the corresponding homology class in $\mathbb{R}^2$. We shall see that this slope is constrained to lie in a narrow band above the irrational value $\log_2 3$.

2.2 The Exact Identity for ρ

Proposition 2.4 (Decomposition of the winding ratio)**.** *Let $n \geq 2$ with terminating Collatz trajectory, and let $n_1, n_2, \ldots, n_{\nu_3}$ be the successive odd values encountered in the orbit (i.e., the values at which $3n_i + 1$ is applied). Then:*

$$\rho(n) = \log_2 3 + \frac{\log_2 n}{\nu_3(n)} + \frac{1}{\nu_3(n)} \sum_{i=1}^{\nu_3(n)} \log_2\Big(1 + \frac{1}{3n_i}\Big). \tag{5}$$

Proof. The trajectory starts at n and ends at 1. Each even step divides by 2, and each odd step multiplies by 3 and adds 1. The total multiplicative effect is:

$$1 = n \cdot \frac{1}{2^{\nu_2}} \cdot \prod_{i=1}^{\nu_3} \frac{3n_i + 1}{n_i} = n \cdot \frac{3^{\nu_3}}{2^{\nu_2}} \cdot \prod_{i=1}^{\nu_3} \Big(1 + \frac{1}{3n_i}\Big). \tag{6}$$

(Note: the factor $\frac{3n_i+1}{n_i}$ accounts for the odd step $n_i \mapsto 3n_i + 1$, and the subsequent halvings contribute $2^{-\nu_2}$ cumulatively.)

Taking $\log_2$ of both sides:

$$0 = \log_2 n - \nu_2 + \nu_3 \log_2 3 + \sum_{i=1}^{\nu_3} \log_2\Big(1 + \frac{1}{3n_i}\Big). \tag{7}$$

Solving for ν_2 and dividing by ν_3:

$$\frac{\nu_2}{\nu_3} = \log_2 3 + \frac{\log_2 n}{\nu_3} + \frac{1}{\nu_3} \sum_{i=1}^{\nu_3} \log_2\Big(1 + \frac{1}{3n_i}\Big). \tag{8}$$

□

Corollary 2.5 (Strict positivity of the deviation)**.** *Define the* winding deviation

$$\delta(n) := \rho(n) - \log_2 3. \tag{9}$$

Then $\delta(n) > 0$ for all $n \geq 2$ with terminating trajectory.

Proof. Both correction terms in (5) are strictly positive: $\log_2 n > 0$ for $n \geq 2$, and $\log_2(1 + 1/(3n_i)) > 0$ for all $n_i \geq 1$. □

The Collatz conjecture as a winding constraint

The equilibrium line $\nu_2 = (\log_2 3)\,\nu_3$ is a **one-sided barrier**: every terminating trajectory has its endpoint strictly above this line in the (ν_2, ν_3)-plane. The Collatz conjecture, in winding number language, asserts that *every lattice walk originating at the origin eventually terminates above the irrational line of slope* $\log_2 3$. A non-terminating trajectory would correspond to a walk whose running ratio $\rho(t)$ approaches $\log_2 3$ from below without ever committing to the upper half-plane.

Remark 2.6 (Decay of the deviation)**.** For "typical" trajectories, the odd values n_i are of order n^c for some $c > 0$, so $\log_2(1 + 1/(3n_i)) \approx 1/(3n_i \ln 2) \to 0$. The dominant contribution to δ is thus $\log_2(n)/\nu_3$. Since ν_3 grows roughly as $\log(n)/\log(4/3)$ for typical n (Terras [3]), the deviation δ remains bounded below by a positive constant for any given n but decreases as ν_3^{-1} for fixed n-size across the population. This $1/\nu_3$ envelope is clearly visible in computational experiments (Section 8).

3 Eigenbasis Coordinates and the Anosov Structure

3.1 The Syracuse Function and Step Sizes

The *Syracuse function* $S : \mathbb{Z}^+_{\mathrm{odd}} \to \mathbb{Z}^+_{\mathrm{odd}}$ is the first-return map of the Collatz iteration to the odd integers:

$$S(n) := \frac{3n+1}{2^{v_2(3n+1)}}, \tag{10}$$

where $v_2(m) := \max\{k : 2^k \mid m\}$ is the 2-adic valuation. One application of S performs exactly one odd step and $k_i := v_2(3n_i+1)$ even steps, so the winding pair advances by the vector $(k_i, 1)$ in the (ν_2, ν_3)-plane.

Definition 3.1 (Syracuse step sequence)**.** The *Syracuse step sequence* of a trajectory from n to 1 is the sequence $\mathbf{k}(n) = (k_1, k_2, \ldots, k_{\nu_3})$ where $k_i = v_2(3n_i + 1)$ is the number of halvings following the i-th odd step.

Proposition 3.2 (Heuristic step distribution)**.** *Under the heuristic assumption that the values $3n_i + 1$ are equidistributed modulo powers of 2 (the "random walk" hypothesis of Lagarias [1]):*

$$P(k_i = k) = 2^{-k}, \quad k \geq 1. \tag{11}$$

The expected step size is $\langle k \rangle = \sum_{k=1}^{\infty} k \cdot 2^{-k} = 2.$

Verification

Over 8,715,141 Syracuse steps collected from 200,000 random trajectories with starting values in $[2, 10^6]$, the empirical mean step size is $\langle k \rangle_{\mathrm{emp}} = 1.9914$, confirming (11) to three significant figures.

3.2 Eigenbasis Coordinates

The Syracuse step structure motivates a change of coordinates on the (ν_2, ν_3)-plane.

Definition 3.3 (Anosov eigenbasis)**.** Define coordinates (u, v) on $\mathbb{R}^2$ by:

$$u := \nu_2 - (\log_2 3)\, \nu_3, \tag{12}$$

$$v := \frac{\nu_2 + (\log_2 3)\, \nu_3}{\sqrt{1 + (\log_2 3)^2}}, \tag{13}$$

where u measures transverse displacement from the equilibrium line and v measures position along it.

In these coordinates, one Syracuse step maps:

$$u \mapsto u + (k_i - \log_2 3), \tag{14}$$

$$v \mapsto v + \frac{k_i + \log_2 3}{\sqrt{1 + (\log_2 3)^2}}. \tag{15}$$

The dynamics in u is a random walk with step sizes $\Delta u_i = k_i - \log_2 3$:

- If $k_i = 1$ (happens with probability $\approx 1/2$): $\Delta u = 1 - \log_2 3 \approx -0.585$. **Contraction.**
- If $k_i = 2$ (probability $\approx 1/4$): $\Delta u = 2 - \log_2 3 \approx +0.415$. **Expansion.**
- If $k_i \geq 3$ (probability $\approx 1/4$): $\Delta u \geq 1.415$. **Strong expansion.**

Proposition 3.4 (Mean drift). *The heuristic mean transverse drift per Syracuse step is:*

$$\langle \Delta u \rangle = \langle k \rangle - \log_2 3 = 2 - \log_2 3 \approx 0.415. \tag{16}$$

Remark 3.5 (Connection to orbit growth and decay). The sign of Δu_i determines the local growth behavior. When $k_i = 1$ (contraction), the Syracuse map sends $n_i \mapsto (3n_i+1)/2 \approx 1.5\, n_i$, so the orbit grows. When $k_i \geq 2$ (expansion in u), the orbit shrinks on average: $n_i \mapsto (3n_i + 1)/2^{k_i}$. The names "contraction" and "expansion" refer to the transverse coordinate u, not the orbit value—they are *opposite* to the behavior of n itself. This inversion is because u measures *excess halvings* over the equilibrium rate.

3.3 Pseudo-Anosov Analogy

The eigenbasis decomposition reveals structural parallels with Anosov diffeomorphisms.

Definition 3.6 (Arnold's cat map). The linear automorphism of $\mathbb{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$ induced by

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \tag{17}$$

stretches by $\lambda = \phi = (1 + \sqrt{5})/2$ along the unstable eigenvector and compresses by $1/\phi$ along the stable eigenvector.

The Collatz dynamics in the (ν_2, ν_3)-plane exhibits analogous behavior:

Feature	**Anosov (cat map)**	**Collatz dynamics**
Phase space	$\mathbb{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$	$(\nu_2, \nu_3) \in \mathbb{Z}^2_{\geq 0}$
Unstable direction	Eigenvector of slope ϕ	Slope $\log_2 3 \approx 1.585$
Stable direction	Eigenvector of slope $-1/\phi$	Slope $-1/\log_2 3 \approx -0.631$
Stretch factor	$\phi \approx 1.618$	Effective: 3/2 (per odd step)
Map type	Linear, uniform	Piecewise, non-uniform
Markov property	Exact (finite partition)	Violated (short-range correlations)

Remark 3.7 (Thurston classification). Thurston's classification [8] of surface homeomorphisms into periodic, reducible, and pseudo-Anosov types provides the correct framework. The Collatz map on the (ν_2, ν_3) torus (reduced modulo k) is not a single linear map but a piecewise map with different behavior on the even/odd branches. The branch locus introduces singularities in the stable and unstable foliations. In Thurston's terminology, this makes the Collatz dynamics *pseudo-Anosov* rather than Anosov: the foliations are measured foliations with finitely many singularities (the prong-type singularities at parity transitions).

4 The Mapping Torus and 3-Manifold Structure

Definition 4.1 (Mapping torus). Let $f : S \to S$ be a homeomorphism of a surface. The *mapping torus* is the 3-manifold

$$M_f := \frac{S \times [0,1]}{(x,0) \sim (f(x),1)}. \tag{18}$$

When f is pseudo-Anosov on $\mathbb{T}^2$, the mapping torus carries *Sol geometry*, one of Thurston's eight 3-dimensional geometries.

For the Collatz dynamics, consider the induced map T_C on the torus $(\mathbb{Z}/k\mathbb{Z})^2$ at modular level k. The mapping torus M_{T_C} fibers over S^1 with fiber $(\mathbb{Z}/k\mathbb{Z})^2$.

Observation 4.2 (Solenoid as inverse limit)**.** The 2-adic solenoid

$$\mathrm{Sol}_2 := \varprojlim\left(\mathbb{R}/\mathbb{Z} \xleftarrow{\times 2} \mathbb{R}/\mathbb{Z} \xleftarrow{\times 2} \cdots\right) \tag{19}$$

is the natural ambient space for the Collatz map, which involves both multiplication by 3 and division by powers of 2. The solenoid is locally homeomorphic to $\mathbb{R} \times \mathbb{Z}_2$ (a line cross a Cantor set). Paths on Sol_2 project to paths on each circle in the inverse system, yielding a coherent system of winding numbers.

The *Collatz solenoid* would be the inverse limit of the mapping tori M_{T_C} at all modular levels k, encoding the full winding number structure.

5 The First-Passage Problem

The transverse coordinate $u(t)$ along a trajectory defines a stochastic process indexed by step number t. By Corollary 2.5, $u(\sigma(n)) > 0$ at termination. But the *running* value $u(t)$ may go negative during the trajectory.

Definition 5.1 (Transverse excursion)**.** The *minimum transverse displacement* of the trajectory from n is

$$u_{\min}(n) := \min_{0 \le t \le \sigma(n)} u(t). \tag{20}$$

A trajectory has a *sub-equilibrium excursion* if $u_{\min}(n) < 0$.

Sub-equilibrium excursions are generic

Among 200,000 trajectories with starting values in $[2, 10^6]$:

- 171,650 (85.8%) have $u_{\min}(n) < 0$.
- The minimum reaches as low as $u_{\min} \approx -17.5$.
- The distribution of $u_{\min}$ has a sharp peak near -1 and a long left tail.

Two-phase dynamics

Collatz trajectories exhibit a characteristic two-phase structure in the eigenbasis:

1. **Excursion phase**: $u(t)$ drifts *downward* (the orbit value n is growing; odd steps dominate the balance). This corresponds to the orbit climbing to its peak altitude.
2. **Descent phase**: $u(t)$ drifts *upward* (halvings dominate; the orbit is collapsing toward 1).

The transition between phases—the *first-passage time* at which $u(t)$ commits to staying above the equilibrium—is the dynamically critical event. Trajectories with long stopping times (like $n = 837{,}799$ with 524 steps) are precisely those where this transition occurs late.

Remark 5.2. The Collatz conjecture in first-passage language: *the random walk $u(t)$ with drift $\langle \Delta u \rangle \approx 0.415$ per Syracuse step is transient to $+\infty$ for every starting value.* Since the drift is positive, the walk is heuristically transient; the difficulty lies in the arithmetic dependence of step sizes k_i on the orbit values, which could in principle produce persistent negative excursions for specific starting numbers.

6 Arithmetic Structure: Torus Residues

Reducing the winding pair modulo k projects the dynamics onto the finite torus $(\mathbb{Z}/k\mathbb{Z})^2$. The resulting density patterns reveal arithmetic constraints imposed by the $3n + 1$ rule.

Definition 6.1 (Torus residue distribution)**.** For modular level $k \geq 2$, define the distribution

$$\mu_k(a, b) := \#\{n \in [2, N] : \nu_2(n) \equiv a,\ \nu_3(n) \equiv b \pmod{k}\} \tag{21}$$

on $(\mathbb{Z}/k\mathbb{Z})^2$.

Forbidden and preferred residue classes

At modular level $k = 3$ (computed for $N = 10^6$):

1. The cell $(\nu_2 \bmod 3, \nu_3 \bmod 3) = (1, 1)$ is essentially *empty*.
2. The cell $(0, 0)$ is the most heavily populated.
3. The distribution is far from uniform: the density ratio between the most and least populated cells exceeds 10^3.

At level $k = 12 = \operatorname{lcm}(3, 4)$, a pronounced checkerboard pattern emerges, aligned with the foliation directions of slope $\log_2 3$ and $-1/\log_2 3$.

Proposition 6.2 (Origin of the mod 3 constraint)**.** *The forbidden cell at $(\nu_2, \nu_3) \equiv (1, 1) \pmod{3}$ is a consequence of the multiplicative balance equation (6). Since $3^{\nu_3} \equiv 0 \pmod{3}$ for $\nu_3 \geq 1$, the equation $2^{\nu_2} = n \cdot 3^{\nu_3} \cdot \prod(1 + 1/(3n_i))$ constrains $2^{\nu_2} \pmod{3}$. Since $2 \equiv -1 \pmod{3}$, we have $2^{\nu_2} \equiv (-1)^{\nu_2} \pmod{3}$, imposing a parity constraint on ν_2 that depends on* $n \bmod 3$.

Remark 6.3 (Significance of mod 24)**.** At modular level $k = 24$, the torus residue distribution shows prominent diagonal banding aligned with the equilibrium foliation. Note that $24 = \operatorname{lcm}(3, 8)$, where the factor of 3 arises from the $3n + 1$ rule and $8 = 2^3$ captures the first three bits of 2-adic structure relevant to the Syracuse step sequence. Whether this is purely combinatorial or reflects deeper structure (cf. the ubiquity of 24 in modular mathematics: $\eta^{24} = \Delta$, $c_{\text{bos}} = 24 + 2$, $|\text{Leech}| = 24$) remains an open question.

7 Step Correlations and the Markov Property

A genuine Anosov diffeomorphism admits a Markov partition, making the symbolic dynamics a shift of finite type. We test whether the Collatz dynamics has this property.

Definition 7.1 (Autocorrelation of Syracuse steps)**.** For a trajectory with Syracuse step sequence $\mathbf{k} = (k_1, \ldots, k_{\nu_3})$, define

$$C(\ell) := \frac{\langle (k_i - \bar{k})(k_{i+\ell} - \bar{k}) \rangle}{\operatorname{Var}(k_i)}, \quad \ell \geq 0, \tag{22}$$

where $\bar{k}$ is the sample mean.

Short-range correlations

For the trajectory from $n = 837{,}799$ ($\nu_3 = 191$ Syracuse steps):

1. $C(1) \approx 0.15$, significantly above the 95% white-noise confidence interval of $\pm 1.96/\sqrt{191} \approx \pm 0.14$.
2. $C(\ell)$ decays to noise by $\ell \approx 5$.
3. The joint distribution $P(k_i, k_{i+1})$ deviates from the product $P(k_i) \cdot P(k_{i+1})$, showing concentration along a diagonal structure.

Proposition 7.2 (Origin of correlations)**.** *The lag-1 correlation follows from elementary number theory. If n_i is odd and $k_i = v_2(3n_i + 1)$, then $n_{i+1} = (3n_i + 1)/2^{k_i}$. The residue class of n_{i+1} modulo small powers of 2 is determined by $n_i \bmod 2^{k_i+j}$ for small j, which constrains $k_{i+1} = v_2(3n_{i+1} + 1)$. Explicitly, n_{i+1} is odd (by construction), and*

$$3n_{i+1} + 1 = 3 \cdot \frac{3n_i + 1}{2^{k_i}} + 1 = \frac{9n_i + 3 + 2^{k_i}}{2^{k_i}}. \tag{23}$$

The 2-adic valuation of the numerator depends on $n_i \bmod 2^{k_i+2}$, creating a correlation between k_i and the residue class that determines k_{i+1}.

Remark 7.3 (Non-uniform hyperbolicity)**.** The presence of short-range correlations means the Collatz dynamics does *not* admit a simple Markov partition in the Syracuse step alphabet. The correct dynamical category is *non-uniformly hyperbolic* in the sense of Pesin theory: the Lyapunov exponents

$$\lambda_u = \lim_{N \to \infty} \frac{1}{N} \sum_{i=1}^{N} \log |k_i - \log_2 3|, \qquad \lambda_s = -\lambda_u \tag{24}$$

exist and are nonzero (positive and negative respectively), but the hyperbolicity is non-uniform—it varies from orbit to orbit and can be transiently reversed. This is the precise obstruction to proving the Collatz conjecture via purely ergodic-theoretic methods.

8 Computational Results

All computations were performed for starting values $n \in [2, 10^6]$ (999,999 trajectories). Summary statistics:

Quantity	Value	Note
ρ range	$[1.687, 21.000]$	Minimum at n with many odd steps
ρ mean	2.1741	Overshoots $\log_2 3 = 1.5850$
ρ std dev	0.4521	
$\delta > 0$ for all n?	**Yes**	Confirms Corollary 2.5
Mean Syracuse step $\langle k \rangle$	1.9914	Predicted: 2.0000
Fraction with $u_{\min} < 0$	85.8%	Sub-equilibrium excursions are generic
Deepest excursion	$u_{\min} \approx -17.5$	
Autocorrelation $C(1)$	≈ 0.15	Significant ($>$ 95% CI)

Remark 8.1 (Extreme deviators)**.** The largest deviations δ occur for numbers that are nearly powers of 2: e.g., $n = 698{,}880 = 2^{10} \cdot 683 - 128$ has $\nu_3 = 1$ and $\nu_2 = 21$, giving $\rho = 21.0$. These are "trivial" outliers—the trajectory encounters only one odd step before collapsing through a long chain of halvings. The dynamically interesting large deviations occur at moderate ν_3 values ($\nu_3 \sim 10$–50) where the quantization of $\rho = \nu_2/\nu_3$ to rational values with small denominator creates a discrete "Farey sequence" structure in the winding spectrum.

9 Summary and Open Questions

9.1 What is established

1. The winding pair (ν_2, ν_3) defines a homology class in $H_1(\mathbb{T}^2; \mathbb{Z}) \cong \mathbb{Z}^2$ for each Collatz trajectory (Definition 2.2).

2. The winding ratio decomposes exactly as $\rho = \log_2 3 +$(positive corrections), proving $\delta > 0$ for all terminating trajectories (Proposition 2.4, Corollary 2.5).

3. The Syracuse step sizes k_i follow $P(k) \approx 2^{-k}$ with empirical mean $1.9914 \approx 2$ (Proposition 3.2), confirming the Lagarias heuristic [1].

4. The transverse coordinate $u(t)$ is a biased random walk with drift $\langle \Delta u \rangle \approx 0.415 > 0$ (Proposition 3.4).

5. Consecutive Syracuse steps exhibit significant short-range correlations ($C(1) \approx 0.15$), ruling out a simple Markov partition (Observation 7, Proposition 7.2).

6. The torus residue distribution μ_k at level $k = 3$ has forbidden cells, traceable to the mod 3 arithmetic of the $3n + 1$ rule (Observation 6, Proposition 6.2).

9.2 Open questions

1. **Markov approximation.** Can the correlated step process (k_i) be approximated by a Markov chain of finite order m on the alphabet of 2-adic residue classes? If so, what is the minimal m, and does the resulting shift space have a tractable zeta function?

2. **First-passage statistics.** Let $\tau(n)$ be the first time t at which $u(t') > 0$ for all $t' \geq t$ (the "commitment time"). What is the distribution of $\tau(n)$ as a function of n? Is $\tau(n)/\sigma(n)$ concentrated, and does it correlate with the stopping time?

3. **Geometry of the mapping torus.** Does the mapping torus M_{T_C} at any finite modular level carry a geometric structure in Thurston's sense? For level $k = 3$, the forbidden cell suggests that T_C is not surjective on $(\mathbb{Z}/3\mathbb{Z})^2$, which would obstruct the construction.

4. **Solenoid realization.** Can the full Collatz dynamics be realized as a continuous map on the $(2, 3)$-solenoid $\varprojlim(\mathbb{R}/\mathbb{Z} \xleftarrow{\times 6} \mathbb{R}/\mathbb{Z} \xleftarrow{\times 6} \cdots)$? If so, the winding numbers at each level form a pro-finite invariant, and the Collatz conjecture becomes a statement about the non-existence of certain closed orbits in the solenoid.

5. **Connection to mod 24 structure.** Is the diagonal banding in μ_{24} purely combinatorial ($24 = \operatorname{lcm}(3, 8)$), or does it reflect deeper modular structure analogous to the role of 24 in the theory of modular forms?

6. **Negative deviation.** While $\delta(n) > 0$ at termination, the running ratio $\rho(t)$ dips below $\log_2 3$ for 85.8% of trajectories. Can one bound the depth $u_{\min}(n)$ as a function of n, and does a uniform lower bound exist?

10 Profinite Compatibility of the Winding Number

The solenoid formulation of §?? predicts that the winding pair $(\nu_2(n), \nu_3(n))$ defines a consistent element of the profinite group × — concretely, that the torus residue distributions μ_k at different modular levels are related by the marginal condition dictated by the inverse limit structure. We test this prediction computationally at all levels $k = 2^a \cdot 3^b$ with $0 \le a, b \le 4$.

10.1 The Compatibility Condition

Definition 10.1 (Marginal consistency)**.** Let $k \mid k'$. The natural projection $\pi_{k',k} : (\mathbb{Z}/k'\mathbb{Z})^2 \to (\mathbb{Z}/k\mathbb{Z})^2$ sends $(a', b') \mapsto (a' \bmod k,\ b' \bmod k)$. The distributions μ_k and $\mu_{k'}$ are *compatible* if

$$\mu_k(a,b) \;=\; \sum_{\substack{a' \equiv a \pmod{k} \\ b' \equiv b \pmod{k}}} \mu_{k'}(a',b') \qquad \text{for all } (a,b) \in (\mathbb{Z}/k\mathbb{Z})^2. \tag{25}$$

If (25) holds for every pair $k \mid k'$ in a cofinal subset of the divisibility poset, then the system $\{\mu_k\}_{k\ge1}$ defines a measure on the inverse limit $\varprojlim(\mathbb{Z}/k\mathbb{Z})^2$.

Remark 10.2. For distributions derived from integer-valued winding pairs, the marginal condition (25) is tautological: $\mu_k(a,b) = \#\{n : \nu_2(n) \equiv a,\ \nu_3(n) \equiv b \pmod{k}\}$, and regrouping the finer residues modulo k' by their classes modulo k recovers μ_k exactly. The computational verification thus serves two purposes: (i) validating the correctness of the implementation across all 25 levels and 200 divisibility pairs, and (ii) confirming that the data infrastructure supports the profinite interpretation without hidden inconsistencies.

10.2 Computation

The profinite compatibility test was originally performed in Python at $N = 2 \times 10^6$. To distinguish genuine forbidden cells from finite-N artifacts and to produce high-statistics entropy curves, we reimplemented the computation in C (`profinite_compat.c`) using a streaming accumulation architecture: for each $n \in [2, N]$, the winding pair (ν_2, ν_3) is computed via Collatz iteration, then immediately accumulated into all 25 distribution grids. No per-trajectory storage is required; total memory is ∼19 MB regardless of N.

Profinite compatibility at $2^a \cdot 3^b$ levels

For $N = 10^9$ starting values (999,999,999 trajectories, mean length 203.2 steps, $\nu_2 \le 616$, $\nu_3 \le 370$), we computed μ_k at all 25 levels

$$k \;\in\; \left\{2^a \cdot 3^b : 0 \le a, b \le 4\right\}, \tag{26}$$

ranging from $k = 1$ to $k = 1296$. For every divisibility pair $k \mid k'$ within this set (200 pairs in total), the marginal condition (25) was verified with

$$\max_{(a,b)\in(\mathbb{Z}/k\mathbb{Z})^2} \left|\mu_k(a,b) - (\pi_{k',k})_*\mu_{k'}(a,b)\right| \;=\; 0 \tag{27}$$

in every case. All 200 tests pass with exact integer agreement. Wall time: 222 s on a single core.

The profinite winding number is well-defined

The system of distributions $\{\mu_k\}_{k=2^a3^b}$ satisfies the inverse limit compatibility condition at all tested levels. Consequently, the profinite winding number

$$\hat{w}(n) \;:=\; \big(\nu_2(n) \bmod 2^a3^b,\;\; \nu_3(n) \bmod 2^a3^b\big)_{a,b\geq 0} \;\in\; \mathbb{Z}_2 \times \mathbb{Z}_3 \tag{28}$$

is empirically confirmed as a consistent profinite invariant of Collatz trajectories. The solenoid formulation $\Sigma_{2,3} \cong (\mathbb{R} \times \mathbb{Z}_2 \times \mathbb{Z}_3)/\mathbb{Z}$ is supported as the correct algebraic home for the winding number data.

10.3 Structural Observations

Beyond the tautological consistency, the distributions μ_k reveal structural phenomena as the level k increases. The $500\times$ increase in sample size (from $N = 2 \times 10^6$ to $N = 10^9$) substantially sharpens these observations.

10.3.1 Support stabilisation

Finite support in the profinite limit

The number of occupied cells in $(\mathbb{Z}/k\mathbb{Z})^2$ stabilises:

k	2	6	12	24	36	72	144	≥ 144
Nonzero cells	4	36	144	576	1,296	3,715	4,089	4,089

For $k \geq 144$, exactly 4,089 of the k^2 cells are occupied. This stabilisation occurs because the winding pairs satisfy $\nu_2(n) \leq 616$ and $\nu_3(n) \leq 370$ for all $n \in [2, 10^9]$, so once k exceeds the data range, residue classes coincide with the winding pairs themselves.
Compared to $N = 2 \times 10^6$ (where the saturation count was 1,576 cells at $k \geq 72$), the $500\times$ increase in sample size reveals 2,513 additional occupied cells. At each fixed level $k \leq 36$, all k^2 cells are now filled, whereas the earlier computation showed empty cells already at $k = 36$.

Remark 10.3. At fixed N, the support is finite. But as $N \to \infty$, the support of μ_k fills a larger fraction of $(\mathbb{Z}/k\mathbb{Z})^2$ at each fixed k. The key question is *which* cells remain empty in the $N \to \infty$ limit. The $500\times$ scale increase already filled all cells at $k \leq 36$ that were previously empty, indicating that the zero cells observed at $N = 2 \times 10^6$ for moderate k were finite-sample artifacts rather than genuine obstructions.

Remark 10.4 (Correction: $\mu_3(1,1)$ is not forbidden)**.** The earlier computation (at $N = 2\times10^6$) reported the cell $(\nu_2, \nu_3) \equiv (1,1) \pmod 3$ as essentially empty. The high-statistics computation reveals this to be incorrect: $\mu_3(1,1)$ scales linearly with N at approximately $N/9$ (i.e., near-uniform):

N	$\mu_3(1,1)$	Expected $(N-1)/9$	Ratio
10^6	112,676	111,111	1.014
10^7	1,098,215	1,111,111	0.988
10^8	11,145,935	11,111,111	1.003
10^9	112,253,124	111,111,111	1.010

The ratio $\mu_3(1,1)/((N-1)/9)$ fluctuates within $\sim$1% of unity at all checkpoints. There is no forbidden cell at level $k = 3$ for the step-counting winding numbers. The mod-3 constraint discussed in Proposition 6.2 constrains the *parity* of ν_2 as a function of n mod 3, but does not prevent (ν_2 mod 3, ν_3 mod 3) $= (1, 1)$ from being populated.

10.3.2 Entropy decay

Entropy of μ_k relative to Haar measure

The normalised Shannon entropy

$$\frac{H(\mu_k)}{\log_2(k^2)} := \frac{-\sum_{(a,b)} p_k(a,b)\log_2 p_k(a,b)}{\log_2(k^2)}, \qquad p_k(a,b) := \frac{\mu_k(a,b)}{\sum \mu_k}, \tag{29}$$

decays monotonically from near-uniformity at small k to substantial concentration at large k:

k	2	3	6	12	24	36	72	144	324	1296
$H/H_{\max}$	1.000	1.000	1.000	0.991	0.926	0.845	0.711	0.611	0.526	0.424

Compared to the $N = 2 \times 10^6$ values (which ranged from 0.999 at $k = 6$ to 0.408 at $k = 1296$), the entropy ratios at $N = 10^9$ are systematically higher at every level—the additional data fills more cells, pushing the distribution closer to uniformity. Nevertheless, the qualitative pattern of monotonic decay is preserved.

Increasing departure from Haar measure

If the Collatz orbit measure on the solenoid $\Sigma_{2,3}$ were Haar measure, μ_k would be uniform on $(\mathbb{Z}/k\mathbb{Z})^2$ and $H/H_{\max} = 1$ at every level. The observed monotonic decay quantifies the **increasing departure from Haar measure** as finer profinite structure is resolved. The Collatz dynamics concentrates the orbit measure onto a progressively smaller fraction of the available phase space at each level — exactly the behaviour expected of a hyperbolic dynamical system, where orbits are attracted to a fractal invariant set of measure zero in the ambient space.

10.3.3 Forbidden cell proliferation

Growth of the forbidden region

The fraction of strictly empty cells in $(\mathbb{Z}/k\mathbb{Z})^2$ grows with k. At $N = 10^9$:

k	k^2	Zero cells	Suppressed ($< 1\%$ of exp.)	Zero fraction
3	9	0	0	0.0%
12	144	0	0	0.0%
24	576	0	0	0.0%
36	1,296	0	138	0.0%
72	5,184	1,469	2,006	28.3%
144	20,736	16,647	1,976	80.3%
324	104,976	100,887	1,488	96.1%
1296	1,679,616	1,675,527	614	99.8%

At $N = 10^9$, the first strictly zero cells appear at $k = 48$ (20 cells) rather than at $k = 36$ as in the earlier computation. The scale increase also reveals a *suppressed* population: cells with $0 < \mu_k(a, b) < \text{expected}/100$, numbering up to 2,006 at $k = 72$. These suppressed cells are candidates for genuine obstructions that may approach zero as $N \to \infty$.

Remark 10.5 (Finite-N versus genuine obstructions)**.** The zero-fraction growth is dominated by the finite support effect (Observation 10.3.1): since only 4,089 distinct winding pairs appear for $N = 10^9$, levels with $k^2 \gg 4{,}089$ are necessarily mostly empty. The $500\times$ scale increase from $N = 2 \times 10^6$ to $N = 10^9$ raised the saturation count from 1,576 to 4,089 and filled all zero cells at $k \leq 36$, demonstrating that the empty cells at moderate k in the earlier computation were statistical artifacts. At $k = 3$, all 9 cells are populated within $\sim$1% of the uniform expectation (Remark 10.4).

The *dynamically meaningful* forbidden cells are those that remain empty (or anomalously suppressed) as $N \to \infty$. The 138 suppressed cells at $k = 36$ and the 2,006 suppressed cells at $k = 72$ warrant investigation at still larger N to determine whether they converge to zero or to small positive densities.

10.4 Interpretation and Consequences

The compatibility result establishes the following.

1. **The profinite winding number exists.** For each Collatz trajectory $n \to 1$, the element $\hat{w}(n) \in \mathbb{Z}_2 \times \mathbb{Z}_3$ is a well-defined profinite invariant that encodes the residue of $(\nu_2(n), \nu_3(n))$ at every level $k = 2^a 3^b$ simultaneously.

2. **The solenoid is the correct phase space.** The consistency of the μ_k system with the inverse limit structure supports the identification of the Collatz phase space with $\Sigma_{2,3} \cong (\mathbb{R} \times \mathbb{Z}_2 \times \mathbb{Z}_3)/\mathbb{Z}$. The orbit measure is a well-defined Borel probability measure on this compact group, and the forbidden cells at each level are its support constraints.

3. **The orbit measure is not Haar.** The entropy decay (Observation 10.3.2) demonstrates that the Collatz orbit measure departs increasingly from Haar measure on $\Sigma_{2,3}$ at finer levels. This confirms and extends the multifractal analysis of Chistyakov embeddings, which showed $D_q^{\text{Collatz}} < D_q^{\text{Haar}}$ for $q > 0$.

4. **Hyperbolic concentration.** The monotonic entropy decay is consistent with the hyperbolic structure of the map $F(x, y) = \left(\frac{x-y}{2}, \frac{3y+1}{2}\right)$ on $\mathbb{Z}_2 \times \mathbb{Z}_3$, whose eigenvalues $\lambda_s = 1/2$ and $\lambda_u = 3/2$ produce area contraction by a factor of $3/4$ per step. The orbit measure concentrates along the unstable manifold of F, and the forbidden cells at each level trace the boundary of this concentration.

5. **Scale resolves artifacts.** The increase from $N = 2 \times 10^6$ to $N = 10^9$ eliminated all zero cells at levels $k \leq 36$ and corrected the earlier report of a forbidden cell at $(\nu_2, \nu_3) \equiv (1, 1) \pmod 3$. The distribution at $k = 3$ is near-uniform, and the genuine structural content of μ_k resides in the *entropy decay* and *suppressed-cell patterns* at intermediate levels $k = 36$–72, where the interplay between the arithmetic of $3n + 1$ and the finite winding-number range creates non-trivial concentration.

Remark 10.6 (What compatibility does *not* establish)**.** The compatibility condition is necessary but not sufficient for the full solenoid program. It confirms that the winding pair defines a consistent profinite invariant, but does not address whether the Collatz map T_C extends continuously to $\Sigma_{2,3}$, nor whether the mapping torus M_{T_C} carries a geometric structure in Thurston's sense. Those questions require the explicit construction of F on $\mathbb{Z}_2 \times \mathbb{Z}_3$ and the analysis of its orbit distribution at finite levels — the next steps in the program.

11 The 6-Adic Ordering Experiment

Conjecture **??** predicts that the Collatz-induced map F on $\mathbb{Z}_2 \times \mathbb{Z}_3$ is measurably conjugate to a skew product over the 6-adic odometer $\tau(x) = x + 1$. A direct consequence is that the profinite winding number $\hat{w}(n)$ should be *approximately locally constant* on 6-adic balls: numbers n, n' with $n \equiv n' \pmod{6^m}$ should have similar winding residues ($\nu_2 \bmod K$, $\nu_3 \bmod K$) for appropriate K. We test this prediction and compare against control bases not involved in the Collatz dynamics.

11.1 Experimental Design

1. Compute $(\nu_2(n), \nu_3(n))$ for all $n \in [2, 10^6]$.

2. For each base $b \in \{6, 5, 7, 10\}$ and level m, partition $[2, N]$ into residue classes modulo b^m ("b-adic balls at depth m").

3. Compute the *within-ball variance* of each observable $\nu_2 \bmod K$, $\nu_3 \bmod K$, and the raw ν_2, ν_3, relative to the total variance:
$$R(b, m) := \frac{V_{\text{within}}}{V_{\text{total}}} = 1 - \frac{SS_{\text{between}}}{SS_{\text{total}}}, \tag{30}$$
where $SS_{\text{between}} = \sum_g n_g(\bar{x}_g - \bar{x})^2$ is the standard ANOVA decomposition. Smaller R (equivalently, larger variance reduction $1 - R$) indicates greater local constancy.

4. Compute the mutual information $I(n \bmod b^m;\ \text{observable})$ in bits as an independent, non-parametric measure of dependence.

5. Compare 6-adic results against 5-adic and 7-adic controls *at matched ball size*, not matched level, since $6^m \neq 5^m \neq 7^m$.

Remark 11.1 (The matched-resolution criterion)**.** A comparison at matched level m is misleading: at level $m = 7$, 6-adic balls contain $\lfloor N/6^7 \rfloor = 3$ elements while 5-adic balls at $m = 8$ contain $\lfloor N/5^8 \rfloor = 2$. The correct test compares bases at comparable modulus (hence comparable ball size), so that any reduction in within-ball variance is not an artifact of small-sample concentration.

11.2 Results: Modular Observables

The key observables for testing Conjecture **??** are the modular winding residues $\nu_2 \bmod 12$ and $\nu_3 \bmod 9$, which capture the fiber data of $\hat{w}(n)$ at the first nontrivial CRT level $k = 2^2 \cdot 3 = 12$ and $k = 3^2 = 9$.

No privileged 6-adic local constancy for modular observables

The variance reduction $1 - R(b, m)$ for $\nu_2 \bmod 12$, compared at matched modulus, shows no systematic advantage for base 6 over controls:

Modulus range	6-adic	7-adic	5-adic	Decimal
~8k–17k	6^5: 0.73%	7^5: 1.77%	—	10^4: 1.00%
~16k–47k	6^6: 4.51%	—	5^6: 1.55%	—
~78k–280k	6^7: 28.2%	7^6: 12.0%	5^7: 7.9%	—
~280k–391k	—	—	5^8: 39.0%	—

Identical behaviour obtains for $\nu_3 \bmod 9$ (6^7: 28.1%, 7^6: 11.9%, 5^8: 39.9%). At the finest resolutions, the 5-adic control achieves *larger* variance reduction than the 6-adic base, and the 7-adic control exceeds 6 at intermediate modulus.

Mutual information confirms the negative result

The mutual information $I(n \bmod b^m;\, \nu_2 \bmod 12)$ at matched level:

m	6-adic	2-adic	3-adic	5-adic	7-adic
1	0.0000	0.0000	0.0000	0.0000	0.0000
2	0.0002	0.0000	0.0000	0.0002	0.0002
3	0.0014	0.0001	0.0002	0.0010	0.0024
4	0.0092	0.0001	0.0005	0.0047	0.0188
5	0.0590	0.0003	0.0016	0.0235	0.1372
6	0.4222	0.0005	0.0050	0.1248	—

The 6-adic MI at $m = 5$ (modulus 7,776) is 0.059 bits, while the 7-adic MI at $m = 5$ (modulus 16,807) is 0.137 bits. At matched modulus, 7-adic balls carry *more* information about $\nu_2 \bmod 12$ than 6-adic balls. The 2-adic and 3-adic components individually carry very little information, suggesting the CRT decomposition $\mathbb{Z}_6 \cong \mathbb{Z}_2 \times \mathbb{Z}_3$ does not concentrate the winding residue information.

11.3 Results: Raw Winding Numbers

A qualitatively different picture emerges for the unmodded winding numbers $\nu_2(n)$ and $\nu_3(n)$.

Genuine 6-adic structure in raw winding numbers

For ν_2 (raw) and ν_3 (raw), the 6-adic base captures significant variance from the very first level:

Level	6-adic	5-adic	7-adic	Decimal
$m = 1$	1.18%	0.000%	0.000%	1.18%
$m = 2$	2.37%	0.000%	0.000%	1.18%
$m = 3$	3.56%	0.001%	0.003%	2.37%
$m = 4$	4.80%	0.005%	0.003%	3.62%
$m = 5$	6.53%	0.028%	0.022%	5.56%

At $m = 1$, the six residue classes $n \bmod 6$ explain 1.18% of the total variance in trajectory length, while $n \bmod 5$ and $n \bmod 7$ explain effectively zero. The decimal base matches at $m = 1$ (sharing the factor 2 with 6).

Remark 11.2 (Arithmetic, not dynamical)**.** The raw-ν signal is an elementary arithmetic effect: $n \bmod 2$ determines the first step of the trajectory (halving or tripling), and $n \bmod 3$ influences divisibility patterns in the first few steps. Together, $n \bmod 6$ predicts the *magnitude* of the total stopping time. This is consistent with the well-known dependence of Collatz trajectory length on initial parity and residue mod 3, and does not constitute evidence for a dynamical conjugacy in the sense of Conjecture **??**. The decimal base, which shares the factor 2 with 6, shows comparable early-level reduction, supporting this interpretation.

11.4 Visualisation

Several diagnostic plots reinforce the numerical findings.

1. **Scatter along 6-adic order** (panels (a)–(d) of Figure **??**). Plotting $\nu_2 \bmod 12$ along the 6-adic index does not reveal step-function block structure. The scatter is visually indistinguishable from the natural ordering and 7-adic control.
2. **Zoom to first 6^4 elements** (panels (e)–(f)). At the scale of 1296 elements, vertical lines at 6-adic ball boundaries $(6^1, 6^2, 6^3)$ show no systematic plateaux in $\nu_2 \bmod 12$ or $\nu_3 \bmod 9$.
3. **Ball-level heatmaps** (Figure **??**). Mean ν_2 and ν_3 within 6-adic balls at levels 1–3 show coherent variation by first digit ($n \bmod 6$, the arithmetic effect) but no hierarchical block structure across digits at levels 2–3.
4. **Conditional distributions** (Figure **??**). The distribution of $\nu_2 \bmod 12$ conditioned on $n \bmod 6 = r$ varies across r, but only weakly. The histograms for different 6-adic balls are not concentrated on distinct residues as a step-function conjugacy would require.
5. **2D scatter colored by p-adic digit** (Figure **??**). Coloring points ($\nu_2 \bmod 12$, $\nu_3 \bmod 9$) by $n \bmod 6$ shows no visible clustering. The control ($n \bmod 7$) is equally formless, confirming the absence of structure.

11.5 Interpretation

The winding number is not locally constant on 6-adic balls

At $N = 10^6$, neither the variance reduction test nor the mutual information test detects privileged local constancy of the profinite winding residues ($\nu_2 \bmod K$, $\nu_3 \bmod K$) on 6-adic balls, compared to 5-adic and 7-adic controls. The raw winding numbers $\nu_2(n)$, $\nu_3(n)$ do show genuine dependence on $n \bmod 6$, but this is attributable to elementary arithmetic and does not constitute evidence for odometer conjugacy.

Remark 11.3 (What this does and does not rule out)**.**

1. The experiment tests a *consequence* of Conjecture **??**, not the conjecture itself. The conjugacy φ might exist but involve a highly non-trivial rearrangement of the 6-adic topology, so that $\hat{w} \circ \varphi^{-1}$ (not $\hat{w}$ itself) is locally constant.
2. The observables tested — $\nu_2 \bmod K$ and $\nu_3 \bmod K$ for fixed small K — may not be the correct fiber coordinates. The conjugacy could act nontrivially on both the base (odometer) and fiber directions simultaneously.
3. At $N = 10^6$, the finest testable 6-adic level with meaningful statistics is $m \leq 5$ (ball size ≥ 128). Deeper levels have too few elements per ball. It is conceivable that 6-adic structure emerges only at a scale inaccessible to this experiment.
4. The experiment *does* rule out the simplest version of the conjecture: that $\hat{w}(n)$ is an approximate function of $n \bmod 6^m$ for moderate m. If the odometer conjugacy holds, it must involve substantial distortion of the 6-adic metric.

[Revised directions for odometer conjugacy]

1. **Test in the F-orbit basis.** Rather than testing $\hat{w}(n)$ against $n \bmod 6^m$, compute the orbit of $F : \mathbb{Z}_2 \times \mathbb{Z}_3 \to \mathbb{Z}_2 \times \mathbb{Z}_3$ directly in truncated p-adic arithmetic and test whether *iterates* of F show 6-adic regularity. The conjugacy may be visible only through the dynamical map, not through the winding number alone.
2. **Non-linear 6-adic coordinates.** Seek a measurable function $\varphi : \mathbb{N} \to \mathbb{Z}_6$ such that $\hat{w}(\varphi^{-1}(x))$ is locally constant. This is a regression problem: find the bijection φ that maximises within-ball homogeneity of $\hat{w}$.
3. **Increase N.** With a C implementation (cf. the $38\times$ speedup of `collatz_tree.c`), extend to $N = 10^9$ to access 6-adic levels $m \leq 7$ with ≥ 100 elements per ball.

12 The Parity Subshift of Finite Type

We now study the binary parity sequence associated to each Collatz trajectory, revealing that its symbolic dynamics is governed by the golden mean shift—a classical subshift of finite type [11]—endowed with a non-maximal (sofic) invariant measure.

Definition 12.1 (Parity sequence)**.** For $n \in \mathbb{N}$, the *parity sequence* $\boldsymbol{\sigma}(n) = (\sigma_0, \sigma_1, \ldots, \sigma_{L-1})$ is the binary string recording the parity of each iterate of the Collatz map T:

$$\sigma_t = \begin{cases} 0 & \text{if } T^t(n) \text{ is even (halving step)}, \\ 1 & \text{if } T^t(n) \text{ is odd (tripling step)}. \end{cases} \tag{31}$$

Here $L = \nu_2(n) + \nu_3(n)$ is the total trajectory length.

12.1 The golden mean constraint

Proposition 12.2 (Forbidden bigram)**.** *For every $n \geq 2$, the parity sequence $\boldsymbol{\sigma}(n)$ contains no occurrence of the bigram* 11*.*

Proof. If $\sigma_t = 1$, then $T^t(n)$ is odd and the Collatz rule yields $T^{t+1}(n) = 3\,T^t(n) + 1$, which is even since $3 \cdot (\text{odd}) + 1$ is always even. Hence $\sigma_{t+1} = 0$. □

Corollary 12.3 (Golden mean shift)**.** *The set of all Collatz parity sequences is contained in the* golden mean shift*, the order-*1 *subshift of finite type $X_F \subset \{0,1\}^*$ defined by the single forbidden word $F = \{11\}$. The adjacency matrix of the underlying directed graph is*

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \tag{32}$$

whose spectral radius is the golden ratio $\phi = \frac{1+\sqrt{5}}{2}$.

No higher-order forbidden words beyond 11

For $n \in [2,\, 5 \times 10^8]$ (98,002,908,272 parity symbols), every forbidden $(m{+}1)$-gram at orders $m = 2, 3, 4, 5$ contains 11 as a substring. No new constraints arise beyond the order-1 golden mean rule. The topological entropy $h_{\text{top}} = \log_2 \phi \approx 0.6942$ bits/step is identical at all orders m.

Remark 12.4. This is consistent with the SFT being exactly the golden mean shift and not a higher-order subshift. A proof that no new forbidden words arise at any order would establish that the Collatz parity language is precisely the set of 11-free binary strings—a statement stronger than Proposition 12.2, which guarantees only inclusion.

12.2 The sofic measure and entropy gap

The golden mean shift carries a one-parameter family of Markov measures, parametrized by $p := P(\sigma_{t+1} = 0 \mid \sigma_t = 0)$. The transition matrix takes the form

$$M = \begin{pmatrix} p & 1-p \\ 1 & 0 \end{pmatrix}, \tag{33}$$

since $P(\sigma_{t+1} = 0 \mid \sigma_t = 1) = 1$ by Proposition 12.2. The stationary distribution is $\pi_0 = 1/(2-p)$, $\pi_1 = (1-p)/(2-p)$.

Collatz transition probabilities

From 5×10^8 trajectories:

Context	$P(\to 0)$	$P(\to 1)$	Count
$\sigma_t = 0$	0.50250	0.49750	64,943,346,555
$\sigma_t = 1$	1.00000	0.00000	32,559,561,718

The empirical parameter is $p \approx 0.5025$, giving stationary probability $\pi_1 \approx 0.3322$. This matches the classical heuristic: each odd step is followed on average by $\bar{k} \approx 2$ halvings (cf. Section 8), yielding $\pi_1 = 1/(1+\bar{k}) = 1/3$.

Proposition 12.5 (Parry measure). *The maximal-entropy (Parry) measure on the golden mean shift has transition probabilities*

$$P_{\text{Parry}}(0 \to 0) = \frac{1}{\phi}, \qquad P_{\text{Parry}}(0 \to 1) = \frac{1}{\phi^2}, \tag{34}$$

with stationary probability $\pi_1^{\text{Parry}} = 1/(\phi + 2) \approx 0.2764$ *and entropy* $h_{\text{Parry}} = \log_2 \phi \approx 0.6942$ *bits/step.*

Proof. The Parry measure is the Markov chain whose transition probabilities are $P(i \to j) = A_{ij}\, v_j/(\phi\, v_i)$, where $v = (\phi, 1)^T$ is the right Perron eigenvector of the adjacency matrix (32). Computing:

$$P(0 \to 0) = \frac{1 \cdot \phi}{\phi \cdot \phi} = \frac{1}{\phi}, \quad P(0 \to 1) = \frac{1 \cdot 1}{\phi \cdot \phi} = \frac{1}{\phi^2}, \quad P(1 \to 0) = \frac{1 \cdot \phi}{\phi \cdot 1} = 1.$$

The stationary distribution satisfies $\pi_1 = \pi_0/\phi^2$ and $\pi_0 + \pi_1 = 1$, giving $\pi_1 = 1/(\phi^2 + 1) = 1/(\phi + 2)$ since $\phi^2 = \phi + 1$. □

The Collatz measure is not the Parry measure

The Collatz parity process defines a sofic measure on the golden mean shift that is *not* the maximal-entropy Parry measure. The empirical transition probability $P(0 \to 1) \approx 0.498$ significantly exceeds the Parry value $1/\phi^2 \approx 0.382$, and the stationary odd-step frequency $\pi_1 \approx 0.332$ exceeds the Parry frequency $1/(\phi + 2) \approx 0.276$. The Collatz measure visits odd steps more often than the maximally random walk on the golden mean shift.

The measure-theoretic entropy of the order-1 Collatz Markov approximation is

$$h_\mu = \pi_0 \cdot H(p,\, 1{-}p) + \pi_1 \cdot H(1,\, 0) = \pi_0 \cdot H(p,\, 1{-}p), \tag{35}$$

since state 1 transitions deterministically to 0. With $p \approx 0.5025$ and $\pi_0 \approx 0.6678$:

$$h_\mu \approx 0.6678 \cdot \big(-0.5025 \log_2 0.5025 - 0.4975 \log_2 0.4975\big) \approx 0.6678 \text{ bits/step.} \tag{36}$$

Entropy convergence across Markov orders

Order m	h_{top} (bits/step)	h_μ (bits/step)
1	0.6942	0.6678
2	0.6942	0.6678
3	0.6942	0.6676
4	0.6942	0.6671
5	0.6942	0.6664

The topological entropy remains $\log_2 \phi$ at all orders, confirming the golden mean shift structure. The measure-theoretic entropy h_μ decreases monotonically with order, as higher-order conditioning reveals additional correlations. The entropy gap $\Delta h = h_{\text{top}} - h_\mu \approx 0.028$ at order 5 quantifies the departure from maximal randomness within the golden mean constraint.

12.3 Long-range parity correlations

Definition 12.6 (Parity autocorrelation)**.** The normalized autocorrelation of the parity process is

$$C(k) = \frac{\langle(\sigma_t - \bar{\sigma})(\sigma_{t+k} - \bar{\sigma})\rangle}{\mathrm{Var}(\sigma_t)}, \quad k \geq 0, \tag{37}$$

where the average is taken over all valid pairs within each trajectory and then across all trajectories.

Parity autocorrelation structure

Lag k	$C(k)$	
0	$+1.0000$	
1	-0.4975	$\approx -1/2$
2	$+0.2590$	$\approx +1/4$
3	-0.1287	$\approx -1/8$
4	$+0.0845$	
5	-0.0245	
6–10	$+0.006$ to $+0.020$	persistent oscillations
11–20	± 0.01 to ± 0.05	non-decaying

The first five lags show geometric decay $C(k) \approx (-1/2)^k$, consistent with the dominant eigenvalue $\lambda_1 \approx -0.498$ of the transition matrix (33). Beyond lag 5, however, persistent oscillations of amplitude ~ 0.01–0.05 remain, exceeding the Markov prediction by orders of magnitude.

Markov residuals

Define $\Delta C_m(k) = |C_{\mathrm{emp}}(k) - C^{(m)}_{\mathrm{model}}(k)|$ for an order-m Markov model fitted to the empirical transition matrices. Summing over lags $k = 1, \ldots, 20$:

Order m	$\sum_{k=1}^{20} \Delta C_m(k)$	$\max_k \Delta C_m(k)$
1	0.374	0.050
2	0.365	0.050
3	0.365	0.050
4	0.365	0.050
5	0.359	0.050

Increasing the Markov order from 1 to 5 reduces the total residual by only 4%. The persistent long-range autocorrelation at lags $k \geq 6$ is not captured by any finite-order Markov model on the parity alphabet. This provides quantitative confirmation of the non-Markov character noted qualitatively in Section 7, now in the parity (rather than Syracuse step) representation.

Remark 12.7 (Connection to the Markov approximation question)**.** The residual autocorrelation in Observation 12.3 bears on the first open question of Section 9: whether the correlated step process admits a finite-order Markov approximation. In the parity alphabet, the answer is quantitatively negative—the process has long-memory structure that no finite-order chain can reproduce. The memory likely arises from the arithmetic correlations identified in Proposition 7.2: the 2-adic structure of $3n+1$ creates dependencies that propagate across many parity steps.

12.4 Synthesis

The Collatz parity measure on the golden mean shift

The Collatz parity process, viewed across all starting values $n \in [2,\, 5 \times 10^8]$, exhibits the following structure:

(i) The symbolic dynamics lies in the **golden mean shift** X_F, $F = \{11\}$, with topological entropy $h_{\text{top}} = \log_2 \phi \approx 0.6942$ bits/step.

(ii) The invariant measure is a **sofic measure** with first-order transition parameter $p \approx 0.5025$ and measure-theoretic entropy $h_\mu \approx 0.6678$ bits/step, giving an entropy gap $\Delta h \approx 0.026$.

(iii) The measure is **not Parry**: the odd-step frequency $\pi_1 \approx 0.332 \approx 1/3$ exceeds the Parry value $1/(\phi + 2) \approx 0.276$, reflecting the arithmetic bias of the $3n+1$ map.

(iv) The process is **not finite-order Markov**: persistent long-range correlations at lags $k \geq 6$ resist approximation by Markov chains up to order 5.

Remark 12.8 (Connection to the pseudo-Anosov structure)**.** The golden mean shift is the symbolic dynamics of the simplest pseudo-Anosov map—the cat map $\left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ on $\mathbb{T}^2$, whose dilatation is precisely ϕ. The appearance of this same shift in the Collatz parity process reinforces the pseudo-Anosov analogy developed in Sections 3–4, but with a crucial difference: the Collatz measure is not the natural (Parry) measure of the shift, and it carries long-range correlations that a genuine Markov partition would preclude. The parity SFT captures the *topological* skeleton of the dynamics; the departure from the Parry measure encodes the *arithmetic* content of $3n+1$.

13 Branch Locus on Finite Tori

The profinite compatibility analysis of Section 10 studied the *final* winding pair $(\nu_2(n), \nu_3(n))$ for each trajectory. We now refine this to *per-step* dynamics: at each Collatz step t, the running residues $(\nu_2(t) \bmod k,\, \nu_3(t) \bmod k)$ identify a cell on the finite torus $(\mathbb{Z}/k\mathbb{Z})^2$, and the parity of the current iterate determines which branch was taken. Cells visited by both branches constitute the *branch locus*—the set where the foliation structure of the Collatz map acquires singularities.

13.1 Setup and observables

Definition 13.1 (Per-step branch data)**.** Fix a level $k = 2^a \cdot 3^b$ with $0 \leq a, b \leq 4$. For each trajectory $n \to 1$, initialise running residues $r_2 = r_3 = 0$. At step t:

- If $T^t(n)$ is even (halving step): increment the even-visit count at cell $(r_2, r_3) \bmod k$, then $r_2 \leftarrow r_2 + 1$.
- If $T^t(n)$ is odd (tripling step): increment the odd-visit count at cell $(r_2, r_3) \bmod k$, then $r_3 \leftarrow r_3 + 1$.

A cell is a *branch cell* if both counts are positive, *pure-even* if only the even count is positive, *pure-odd* if only the odd count is positive, and *empty* otherwise.

Definition 13.2 (Per-cell observables)**.** For each non-empty cell $(a, b) \in (\mathbb{Z}/k\mathbb{Z})^2$, define:

$$p_{\text{odd}}(a,b) = \frac{n_{\text{odd}}}{n_{\text{even}} + n_{\text{odd}}}, \tag{38}$$

$$s(a,b) = \frac{p_{\text{odd}}}{1 - p_{\text{odd}}}, \quad \text{(local slope)} \tag{39}$$

$$H(a,b) = -p \log_2 p - (1-p) \log_2(1-p), \quad \text{(binary entropy)} \tag{40}$$

$$\Delta(a,b) = \left|s(a,b) - 1/\log_2 3\right|. \quad \text{(slope deviation)} \tag{41}$$

A branch cell is *singular* if $\Delta(a,b) > \bar{\Delta} + 2\sigma_\Delta$, where $\bar{\Delta}$ and σ_Δ are the mean and standard deviation of Δ over all branch cells at level k.

The local slope $s(a,b)$ should average to $1/\log_2 3 \approx 0.6309$ if the dynamics at that cell tracks the global foliation. Singular cells are those where the local dynamics deviates strongly from the expected foliation slope.

Remark 13.3 (Foliation comparison)**.** The two foliations on $(\mathbb{Z}/k\mathbb{Z})^2$ relevant to the pseudo-Anosov structure (Sections 3–4) are:

Foliation	Slope in (ν_2, ν_3)	Angle
Unstable	$\log_2 3 \approx 1.585$	57.8
Stable	$-1/\log_2 3 \approx -0.631$	−32.2

13.2 Branch fraction and saturation

The computation was performed for all $n \in [2, 10^9]$ ($\sim 2 \times 10^{11}$ total steps) across 25 levels $k = 2^a \cdot 3^b$, $0 \le a, b \le 4$.

Branch fraction progression

k	(a,b)	k^2	Branch	Empty	Br. fraction	Singular
72	$(3,2)$	5,184	5,184	0	1.000	95
81	$(0,4)$	6,561	6,561	0	1.000	188
108	$(2,3)$	11,664	11,664	0	1.000	65
144	$(4,2)$	20,736	16,371	3,706	0.789	127
216	$(3,3)$	46,656	≈16,371	≈29,626	≈0.351	127
1296	$(4,4)$	1,679,616	≈16,371	≈1,662,586	≈0.010	127

All levels $k \le 108$ are fully saturated (100% branch cells) at $N = 10^{10}$. Above $k = 108$, the branch fraction drops sharply as empty cells dominate. For $k \ge 144$, the branch cell count is approximately stable across resolutions ($\sim$16,371 at $N = 10^{10}$, within $\sim$1% across k), consistent with the support stabilisation observed in Section 10. The count grows with N (see Remark 13.7).

13.3 The singular cell anomaly at $k = 81$

The most striking feature of the data is the singular cell count at $k = 81 = 3^4$.

Singular cell density by level

k	Factorisation	Singular	Branch cells	Density
72	$2^3 \cdot 3^2$	95	5,184	1.83%
81	$\mathbf{3^4}$	**188**	**6,561**	**2.86%**
108	$2^2 \cdot 3^3$	65	11,664	0.56%
144	$2^4 \cdot 3^2$	127	16,371	0.78%

The singular cell density at $k = 81$ exceeds that of all other levels, including larger tori. Adding a factor of 4 (passing from $81 = 3^4$ to $108 = 2^2 \cdot 3^3$) *reduces* the singular density by a factor of 5.

13.4 Sublattice analysis: no 3-adic clustering

The initial hypothesis—that singular cells at $k = 81 = 3^4$ cluster on the $\nu_3 \equiv 0 \pmod 3$ sublattice—was tested and decisively rejected.

Uniform distribution on all sublattices

For the 188 singular cells at $k = 81$, the marginal distributions of ν_2 mod m and ν_3 mod m for $m \in \{3, 9, 27\}$ are indistinguishable from uniform:

mod m	$\chi^2(\nu_2)$	$\chi^2(\nu_3)$	df	Expected (uniform)
3	0.01	0.14	2	62.7
9	1.00	0.43	8	20.9
27	14.2	3.3	26	7.0

The joint distribution (ν_2 mod 3, ν_3 mod 3) has $\chi^2 = 0.43$ on 8 degrees of freedom, consistent with perfect uniformity. The singular cells are not attracted to any 3-adic sublattice.

13.5 Singular cell chains along the $\log_3 2$ direction

The spatial structure of singular cells, however, is highly non-trivial.

Connected components

The 188 singular cells at $k = 81$ form 65 connected components (8-connected on the torus):

Component size	Count
37	1
18	3
10	1
7	4
4	1
1 (isolated)	55

There are no size-2 pairs. The 133 cells in components of size ≥ 3 form extended chains.

Chain orientation converges to $\log_3 2$

Principal component analysis of each chain of size ≥ 3 yields a consistent principal direction:

Size	Chain slope	$\lvert\text{slope} - \log_3 2\rvert$	Mean $\Delta(a,b)$
37	0.640	0.009	0.599
18 ($\times$3)	0.640	0.009	0.622
10	0.638	0.007	0.631
7 ($\times$4)	0.640	0.009	0.581

All chains are aligned at slope ≈ 0.640, deviating from $\log_3 2 = 1/\log_2 3 \approx 0.6309$ by less than 0.01. This is neither the unstable foliation slope ($\log_2 3 \approx 1.585$) nor the stable foliation slope ($-1/\log_2 3 \approx -0.631$), but the *positive reflection* of the stable slope: $+\log_3 2$.

Remark 13.4 (Diophantine interpretation)**.** The best rational approximation of $\log_3 2$ with denominator dividing $81 = 3^4$ is $51/81 = 17/27 \approx 0.6296$, with error 1.3×10^{-3}. The chains trace the direction in which $\log_3 2$ is approximated by the 3^4-lattice. The chain lengths (7, 10, 18, 37) may reflect how far one can walk along the $\log_3 2$ direction before the Diophantine approximation error accumulates enough to push p_{odd} back above the singular threshold.

13.6 Even-dominated dynamics at singular cells

Singular cells are almost pure-even

The parity ratio at singular cells is radically different from the global average:

Cell class	Mean p_{odd}	Std. dev.
Global	0.3324	—
Non-singular (k=81)	0.3070	0.060
Singular (k=81)	**0.0595**	0.073

Of the 188 singular cells, 187 (99.5%) have $p_{\text{odd}} < 0.3324$. These are torus positions where trajectories almost exclusively take the even branch (halving), rarely encountering an odd step.

Even tunnels along $\log_3 2$

The singular cells at $k = 81$ form coherent chains of predominantly even-step dynamics, threaded along the $\log_3 2$ direction on the torus. On the pure-3^4 lattice, the even branch $x \mapsto x/2$ has no resonance with the lattice structure ($\gcd(2, 81) = 1$), while the odd branch $x \mapsto 3x + 1$ increments ν_3 by 1 with perfect lattice alignment. The chains mark exactly where the irrationality of $\log_3 2$ creates coherent "tunnels" of predominantly even dynamics—regions where the 3-adic lattice cannot approximate the conjugate winding slope.

13.7 Spatial and foliation-distance analysis

Spatial clustering

Singular cells exhibit strong spatial clustering on the $k=81$ torus:

Statistic	Value
Expected neighbours (random)	0.23
Observed mean neighbours	1.45
Enrichment factor	6.3×

The nearest-neighbour count is 6.3× the random expectation. Half of all singular cells have exactly 2 neighbours, forming the interior of chains.

Enrichment near unstable foliation

Singular cells concentrate near the *unstable* foliation (slope $\log_2 3$):

Distance bin (unstable)	Singular/Total	Enrichment
$[0, 4.3)$	60/1313	1.59×
$[4.3, 8.6)$	47/1311	1.25×
$[8.6, 13.0)$	30/1312	0.80×
$[13.0, 17.3)$	26/1312	0.69×
$[17.3, 21.6)$	25/1312	0.66×

The enrichment decreases monotonically from 1.59× to 0.66× with distance from the unstable foliation. No comparable enrichment is observed with respect to the stable foliation (enrichment factors 0.93–1.12×, essentially flat).

Remark 13.5 (Geometric picture)**.** The singular cells are chains *aligned along* $\log_3 2$ that *concentrate near the* $\log_2 3$ *foliation.* These two directions are not identical: the chain direction (32.6 from the ν_2-axis) differs from the unstable foliation (57.8) by 25.2. The chains thread along the conjugate slope, but preferentially populate the neighbourhood of the unstable foliation.

13.8 Phase transition at the saturation boundary

Two regimes of singular cells

The character of singular cells reverses sharply at the boundary where the branch fraction drops below 1:

Regime 1 ($k \leq 81$, 100% branch fraction):

k	Fact.	Sing.	Max chain	Chain slope	Mean p_{odd}	% below global
48	$2^4 \cdot 3$	26	7	0.673	0.170	100%
54	$2 \cdot 3^3$	44	10	0.732	0.124	97.7%
72	$2^3 \cdot 3^2$	95	24	0.638	0.036	98.9%
81	3^4	188	37	0.640	0.060	99.5%

Regime 2 ($k \geq 108$, $<$ 100% branch fraction):

k	Fact.	Sing.	Max chain	Chain slope	Mean p_{odd}	% below global
108	$2^2 \cdot 3^3$	65	10	0.791	0.686	0.0%
144	$2^4 \cdot 3^2$	127	9	0.760	0.695	0.0%

Phase transition in singular cell character

The two regimes exhibit a complete reversal:

(i) **Below saturation** ($k \leq 81$): Singular cells have very low p_{odd} (0.036–0.170), forming "even tunnels" aligned at slope $\log_3 2$. The chain slope converges to $\log_3 2$ with error < 0.01 at $k = 72$ and $k = 81$.

(ii) **Above saturation** ($k \geq 108$): Singular cells flip to high p_{odd} (0.686–0.695), forming "odd outliers" with shorter chains and slope drifting away from $\log_3 2$.

(iii) The transition coincides with the appearance of empty cells. On a fully saturated torus, the singular set reveals the *intrinsic* foliation singularities of the Collatz dynamics. On a partially empty torus, the singular set instead reflects the *boundary* of the occupied region.

13.9 Diophantine foliation analysis

The rational approximation quality of $\log_3 2$ by fractions $p/3^b$ varies dramatically with b. Only certain exponents yield *record approximations* (strictly improving on all smaller powers of 3):

b	3^b	Best $p/3^b$	Approx. value	$\lvert p/3^b - \log_3 2\rvert$
1	3	2/3	0.66667	3.57×10^{-2}
3	27	17/27	0.62963	1.30×10^{-3}
6	729	460/729	0.63100	7.16×10^{-5}
9	19,683	12419/19683	0.63095	2.08×10^{-5}
11	177,147	111777/177147	0.63093	1.76×10^{-6}

The levels $3^4 = 81$ through $3^5 = 243$ all use the same best approximant 17/27; the next jump in accuracy occurs at $3^6 = 729$, which is 18× more precise. We add $k = 729$ and $k = 19{,}683 = 3^9$

(a further 3.4× improvement, 12419/19683) as additional torus levels to probe this Diophantine structure and measure the scaling of strip width across Diophantine jumps.

Observation 13.6 (Frozen branch count)**.** At $N = 10^{10}$, the number of branch cells is approximately stable across resolutions $k \geq 144$: $k = 144$ has 16,371 branch cells and $k = 729$ has 16,577 (within ∼1%). At $k = 729$ this gives 16,577 branch cells of 531,441 total (branch fraction 3.1%). The branch locus has *finite complexity*: the set of trajectory signatures producing branching behaviour is approximately captured by $k \approx 144$, and larger moduli embed essentially the same set in a sparser grid.

Remark 13.7 (Branch count convergence with N)**.** The "frozen" property refers to approximate stability across k (resolution-independence), not convergence in N (trajectory count). As N increases, rare trajectories visit previously unoccupied cells with both parities, converting pure cells to branch cells. The branch count at $k = 144$ grows substantially with N: 9,415 ($N = 10^8$) → 13,688 ($N = 10^9$) → 16,371 ($N = 10^{10}$), an increase of +74% from 10^8 to 10^{10}. The branch count has **not converged** at $N = 10^{10}$. Nevertheless, the qualitative structure—approximately resolution-independent branch count, finite topological complexity—is robust across all N tested.

Observation 13.8 (Foliation depletion at Diophantine levels)**.** The enrichment of branch cells on the unstable foliation reverses at large k. At $N = 10^{10}$:

k	branch	branch ∩ unstable fol.	enrichment (unstable)	enrichment (stable)
6–72	all	all	1.000	1.000
144	16,371	—	—	—
729	16,577	12	0.045	0.426

At $k \leq 72$ (full saturation), every foliation cell is trivially a branch cell. At $k = 729$, the branch locus is 22× *depleted* on the unstable foliation (enrichment 0.045)—branch cells actively avoid the unstable foliation lines. The stable foliation shows a milder depletion (enrichment 0.426, a 2.3× depletion).

Observation 13.9 (Diagonal band structure at $k = 729$)**.** At $N = 10^{10}$, 19,025 non-empty cells occupy 3.6% of the 729 × 729 torus. All non-empty cells lie on a single coherent diagonal band of width ∼20 cells, aligned at slope $\approx \log_3 2$. Within this band:

- Branch cells (gold) occupy the core (16,577),
- Pure-even cells (893) flank the branch cells on both sides, forming "laminar tunnel walls,"
- Pure-odd cells: 0 (eliminated at $N = 10^{10}$; formerly 41 at $N = 10^9$).

The band is *offset* from the exact unstable foliation by 15–30 cell units in perpendicular distance. The radial density profile from the foliation shows non-empty cells peaking at distance ∼20, not at distance 0.

Observation 13.10 (Foliation shadow)**.** The offset between the non-empty band and the exact foliation can be understood via the Diophantine approximation. At $k = 729$, the best rational approximation $460/729 \approx 0.63100$ exceeds $\log_3 2 \approx 0.63093$ by 7.16×10^{-5}. Over 729 cells, this error accumulates to $729 \times 7.16 \times 10^{-5} \approx 0.052$ wrapping cycles, producing a systematic drift of the trajectory band away from the irrational foliation. The band follows the *rational approximation* $\nu_3 = (460/729)\,\nu_2$, not the irrational foliation $\nu_3 = \log_3 2 \cdot \nu_2$. We call this the *foliation shadow*: trajectories are enslaved to the best rational approximation, creating a narrow corridor that shadows but does not coincide with the true foliation.

Observation 13.11 (Parity distribution at Diophantine levels)**.** The distribution of p_{odd} among branch cells at $k = 729$ is heavily left-skewed relative to $k = 81$, with the bulk of the mass below $p_{\text{odd}} = 0.1$. This confirms that the Diophantine branch cells are even-dominated: the better the rational approximation, the more the dynamics within the foliation shadow are biased toward even (halving) steps.

Diophantine quality controls foliation geometry

The quality of the rational approximation $p/3^b \approx \log_3 2$ determines three geometric features of the branch locus on the 3^b-torus:

(i) **Band width:** Better approximations produce tighter trajectory confinement. At $k = 81$ (error 1.3×10^{-3}), all cells are visited; at $k = 729$ (error 7.2×10^{-5}), only 3.6% of cells are non-empty (at $N = 10^{10}$).

(ii) **Foliation offset:** The trajectory band follows the rational approximation, not the irrational foliation, with the offset proportional to the approximation error.

(iii) **Parity lamination:** Within the band, even-step cells dominate, with branch cells (mixed parity) confined to the core and pure-even cells forming the walls—a laminar structure invisible at saturated levels.

13.10 Parity transition tracking and "11" verification

To directly verify the SFT forbidden bigram "11" (Section 12) at the level of individual torus cells, and to compute per-cell transition valence for the Euler characteristic programme, we extend the branch locus computation with per-cell parity transition grids.

Setup. For each target level $k \in \{81, 108, 144, 729\}$, we track three additional $k \times k$ grids recording the number of *even→even*, *even→odd*, and *odd→even* transitions at each cell (r_2, r_3). A transition is recorded at the cell of the current step, with the transition type determined by the previous and current parities. No *odd→odd* grid is needed: after an odd Collatz step $x \mapsto 3x + 1$, the result is always even, so the next step is necessarily even. Memory cost: $3 \times k^2 \times 8$ bytes per target level, ∼52 MB total.

Observation 13.12 ("11" forbidden bigram verified on torus cells)**.** At $N = 10^{10}$ (2.27×10^{12} total steps), the transition counts at all four target levels satisfy

$$\text{ee} + \text{eo} + \text{oe} = N_{\text{steps}} - N_{\text{trajs}} = 2{,}262{,}503{,}599{,}734$$

exactly, with zero residual. Since the only missing transition type is odd→odd, this confirms oo = 0 at every cell—the "11" forbidden bigram of the parity SFT (Section 12) is verified cell-by-cell on the torus.

Observation 13.13 (Transition fractions and valence)**.** At $N = 10^{10}$, the global transition fractions are ee : eo : oe = 0.3345 : 0.3316 : 0.3339, nearly equal thirds. The slight asymmetry reflects the departure of $p_{\text{odd}} \approx 0.3324$ from the equilibrium value $1/(1 + \log_2 3) \approx 0.387$ at finite N.

The *transition valence* of a cell—the number of distinct transition types (ee, eo, oe) observed—ranges from 0 (empty) to 3 (mixed). At $k = 729$, the valence map reveals internal structure within the foliation shadow: cells in the strip core have valence 3, while cells at the strip edges have valence 1 or 2, with the pattern modulated by the 460/729 rational lattice.

Observation 13.14 ("11" successor analysis)**.** For each odd-visited cell (r_2, r_3) at $k = 729$, we classify the successor cell $(r_2, (r_3 + 1) \bmod k)$—the cell where the trajectory lands after the odd step. At $N = 10^{10}$ (17,470 odd-visited cells):

Successor type	Count	Fraction
branch (both parities)	16,577	94.9%
pure_even	893	5.1%
pure_odd	0	0%
empty	0	0%

The successor of an odd visit is *never* a pure-odd cell, confirming that the "11" constraint maps directly onto the pure-even tunnel walls of the foliation shadow. The 5.1% of successors landing in pure-even cells correspond to the strip boundary, where the odd step shifts r_3 by 1 and pushes the trajectory outside the branch core. This boundary structure manifests as "blue tendrils" extending from the branch strip in the direction of the stable foliation.

Observation 13.15 (Shadow offset spatial structure)**.** The shadow offset $\delta = d_{\mathrm{irr}} - d_{\mathrm{rat}}$ (distance from irrational foliation minus distance from rational foliation) exhibits a crisp sign change across the foliation strip at $k = 729$:

- Cells *above* the irrational foliation (larger ν_2 for given ν_3): $\delta < 0$ (closer to irrational).
- Cells *below* the irrational foliation: $\delta > 0$ (closer to rational).

The boundary between positive and negative offset *is* the irrational foliation line itself. The offset magnitude grows linearly along the strip as $\Delta s{\cdot}r_2/\sqrt{1+s^2}$, where $\Delta s = 460/729 - \log_3 2 \approx 7.16 \times 10^{-5}$, reaching a maximum of ≈ 0.044 cell units at $r_2 = 729$. Since the rational slope exceeds the irrational by $\Delta s > 0$, the rational foliation sits slightly below the irrational in the (ν_3, ν_2) plot, producing the asymmetry: cells above the irrational line are farther from the rational line (and vice versa). The two foliations are at most 0.044 cells apart, so essentially no branch cells sit between them—every cell is unambiguously on one side.

SFT–tunnel correspondence at k

The per-cell transition analysis establishes a direct link between the symbolic dynamics (parity SFT, Section 12) and the geometric structure (foliation shadow, Observation 13.10):

(i) The forbidden bigram "11" is verified at every cell of every target level (implied $\mathrm{oo} = 0$).

(ii) The successors of odd-visited cells are never pure-odd, confirming that the "11" constraint creates the pure-even tunnel walls flanking the branch core.

(iii) The shadow offset cleanly bisects the strip into cells closer to the rational vs. irrational foliation, with the irrational line as the precise dividing surface.

13.11 Synthesis

Branch locus structure on finite tori

The per-step Collatz dynamics on $(\mathbb{Z}/k\mathbb{Z})^2$ for $k = 2^a \cdot 3^b$ $(0 \leq a, b \leq 4)$ and the Diophantine level $k = 3^6 = 729$, computed from up to 10^{10} trajectories (2.27×10^{12} steps) with per-cell parity transition tracking, reveals:

(i) **Universal branching at saturation.** All levels $k \leq 108$ have 100% branch cells at $N = 10^{10}$. Above $k = 108$, the branch count is approximately stable across resolutions (∼16,371 at $k = 144$, ∼16,577 at $k = 729$, within ∼1%). The count grows with N and has not converged (see Remark 13.7).

(ii) **Singular chains along $\log_3 2$.** At the fully saturated levels $k = 72$ and $k = 81$, singular cells form extended chains (up to 37 cells) aligned at slope $\log_3 2 \approx 0.631$ with precision < 0.01. These chains mark coherent regions of predominantly even dynamics.

(iii) **Foliation enrichment and depletion.** At saturated levels ($k \leq 72$), singular cells are 1.6× enriched near the unstable foliation. At the Diophantine level $k = 729$, branch cells are 22× *depleted* on the unstable foliation (enrichment 0.045)—a complete reversal.

(iv) **Foliation shadow.** At $k = 729$, non-empty cells form a single diagonal band of width ∼20 cells at slope $\log_3 2$, offset by 15–30 cell units from the exact foliation. The band follows the best rational approximation $460/729 \approx 0.63100$, not the irrational $\log_3 2$.

(v) **Parity lamination.** Within the foliation shadow, even-step cells dominate: branch cells (mixed parity) form the core, flanked by pure-even walls. The p_{odd} distribution at $k = 729$ is heavily left-skewed (< 0.1).

(vi) **Phase transition.** The singular cell character reverses at the saturation boundary: even-dominated tunnels ($p_{\text{odd}} \approx 0.06$) below, odd-dominated outliers ($p_{\text{odd}} \approx 0.69$) above.

(vii) **No sublattice structure.** Despite $k = 81 = 3^4$ being a pure prime power, the singular cells are uniformly distributed across all 3-adic sublattices.

(viii) **"11" verification.** Per-cell transition grids at target levels $k \in \{81, 108, 144, 729\}$ confirm that oo $= 0$ at every cell (2.27×10^{12} transitions verified). The successors of odd-visited cells at $k = 729$ are never pure-odd (94.9% branch, 5.1% pure-even), directly linking the parity SFT forbidden bigram to the pure-even tunnel walls of the foliation shadow.

(ix) **Shadow offset bisection.** The shadow offset $d_{\text{irr}} - d_{\text{rat}}$ exhibits a crisp sign change at the irrational foliation, bisecting the strip into cells closer to the rational (460/729) vs. irrational ($\log_3 2$) foliation. The offset grows linearly as $\Delta s \cdot r_2/\sqrt{1+s^2}$ with maximum 0.044 cell units, confirming that the two foliations are nearly coincident at $k = 729$.

Remark 13.16 (Irrational foliation passes through the branch strip)**.** A natural question is whether the irrational foliation (the equilibrium line $\nu_3 = \log_3 2 \cdot \nu_2$) is geometrically separated from the branch cells. Direct computation at $k = 729$ shows it is *not*: 180 branch cells lie within 0.5 cell units of the irrational foliation, with the minimum distance ≈ 0. The irrational line passes through the densest part of the branch strip.

The "15–30 cell offset" (Observation 13.8) refers to the peak of the *radial density profile* measured from the *unstable* foliation (slope $\log_2 3$), not the distance from the trajectory band to the irrational line itself. This is consistent with the shadow offset bisection (Observation 13.15), which shows the irrational foliation as the *dividing surface within* the strip.

The mechanism by which trajectories are forced to descend is therefore not geometric exclusion ($A \cap L = \emptyset$, where A is the branch locus and L the equilibrium line). Instead, the pure-even tunnel walls flanking the branch core act as **reflecting boundaries**: when a trajectory approaches the tunnel edge, it encounters pure-even cells that force $n/2$ divisions, pushing the walk $u(t)$ downward. The Baker bound guarantees that these walls persist at every scale—the tunnel has minimum width bounded away from zero—ensuring the reflecting boundary is permanent.

Remark 13.17 (The role of $\log_3 2$ and its Diophantine approximants)**.** The appearance of $\log_3 2 = 1/\log_2 3$ as the chain direction is significant. The two fundamental slopes of the Collatz foliation are $\log_2 3$ (unstable) and $-1/\log_2 3$ (stable). The singular chains are aligned along the *magnitude* of the stable slope, but with positive sign—neither foliation direction, but the conjugate winding ratio $\log_3 2$ that measures the average number of halving steps per tripling step. This is the same quantity that governs the drift along the unstable eigendirection (Section 3).

The Diophantine analysis (Section 13.13.10) confirms this interpretation: at levels where $p/3^b$ is a record approximation to $\log_3 2$, trajectories are enslaved to the rational foliation $\nu_3 = (p/3^b)\,\nu_2$, forming a narrow band that shadows the irrational line. The quality of the approximation controls the band width (tighter $\Rightarrow$ narrower) and the foliation offset (tighter $\Rightarrow$ closer, but never coincident). The sequence of Diophantine best approximants $b = 1, 3, 6, 9, 11, \ldots$ defines a hierarchy of torus levels at which the foliation shadow is maximally constrained, potentially connecting to the continued-fraction expansion of $\log_3 2$ and the Baker-type lower bounds on $|2^m - 3^n|$.

Remark 13.18 (Implementation)**.** The branch locus computation is implemented in `branch_locus.c`, which performs per-step streaming accumulation across 26 torus levels simultaneously (25 standard levels $2^a \cdot 3^b$ with $0 \le a, b \le 4$, plus the Diophantine level $3^6 = 729$). Per-cell parity transition grids (ee, eo, oe) are tracked at four target levels $k \in \{81, 108, 144, 729\}$, adding $\sim$52 MB. Total memory: $\sim$100 MB. The definitive $N = 10^{10}$ run completed in 19.5 hours, processing 2.27×10^{12} steps ($\sim 32 \times 10^6$ steps/s). Per-cell data is written to CSV (`branch_cells.csv`, `branch_transitions.csv`, `branch_shadow.csv`) for offline analysis. Companion plots are generated by `plot_branch_locus.py` (standard levels) and `plot_diophantine.py` (Diophantine and transition analysis, 5 figures).

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